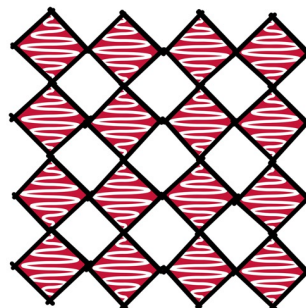




**Northumbria
University**
NEWCASTLE

Chalcogenide Perovskites- a new class of functional materials?

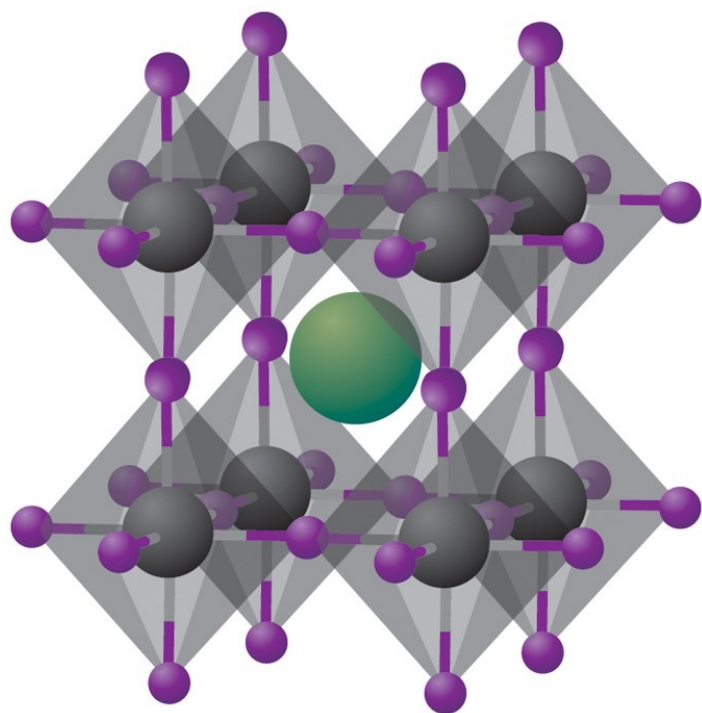


Dr Lucy Whalley

Assistant Professor in Physics

Northumbria University

Perovskites, ABX_3



(In)organic **A**-site

Inorganic
 BX_6
octahedra

oxides

$CaTiO_3$
 $BaTiO_3$
 $LaMnO_3$
....

halides

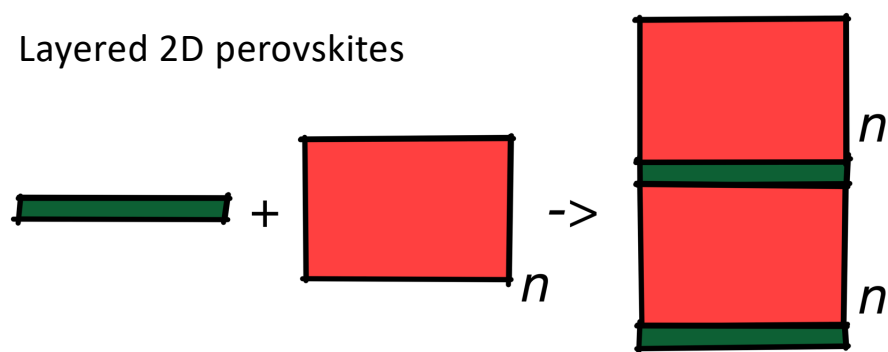
$MAPbI_3$
 $CsPbI_3$
 $(Cs,FA)PbI_3$
 $MAPb(I,Br)_3$
 $(MA,Cs)Pb(I,Br)_3$
 $(FA,MA,Cs)Pb(I,Br)_3$
....

chalcogenides

$BaZrS_3$
 $BaZr(S,Se)_3$
 $Ba(Zr,Ti)S_3$
....

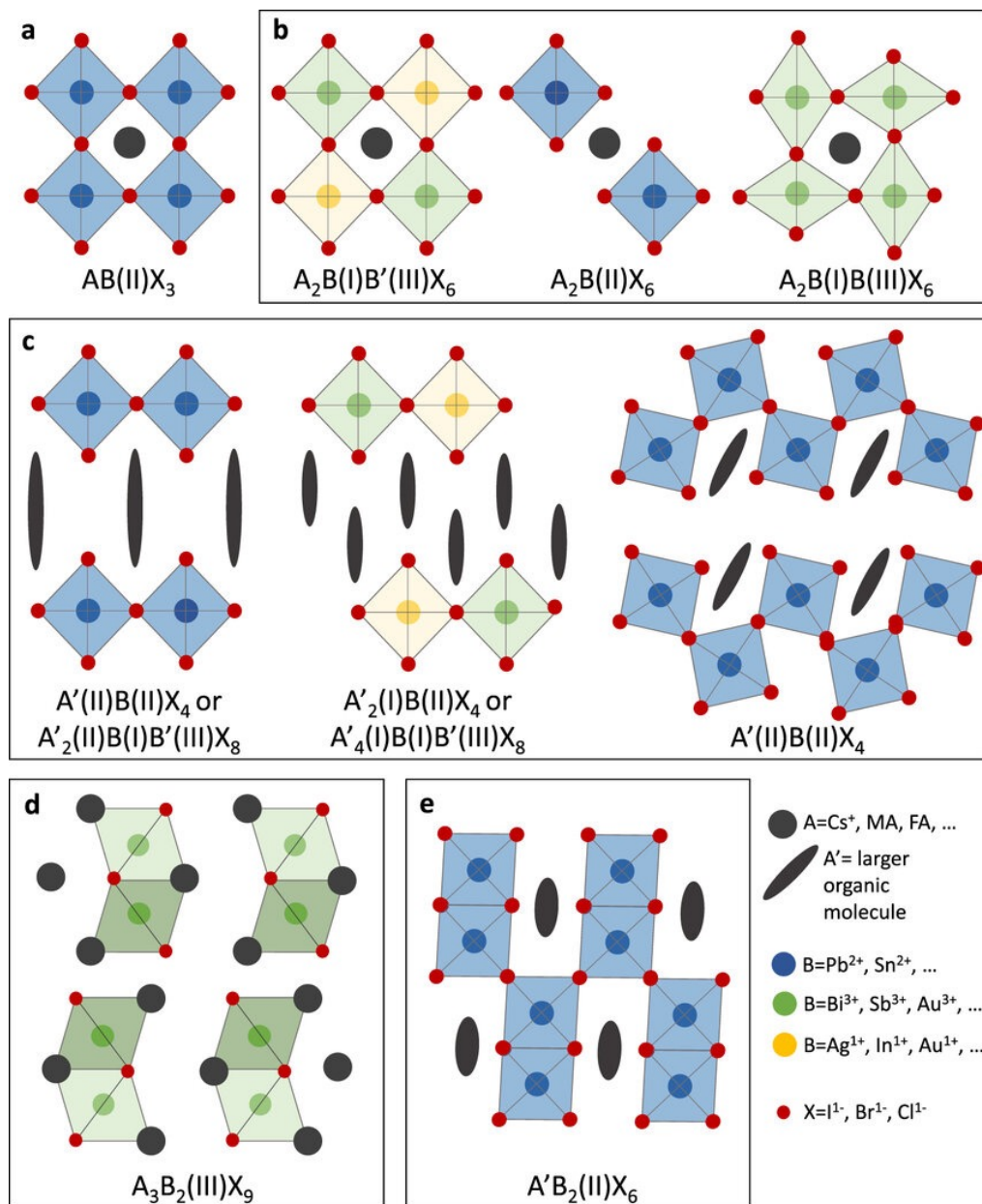
Perovskite-like crystal structures

Layered 2D perovskites



Name	Formula	Example
Ruddlesden Popper	$[A'_2]-[A_{n-1}B_nX_{3n+1}]$	$\text{Ba}_4\text{Zr}_3\text{S}_{10}$
Dion-Jacobsen	$[A']-[A_{n-1}B_nX_{3n+1}]$	$\text{CsLaTa}_2\text{Se}_7$
Aurivillius	$[A'_2X_2]-[A_{n-1}B_nX_{3n+1}]$	$\text{BaBi}_2\text{Nb}_2\text{O}_9$

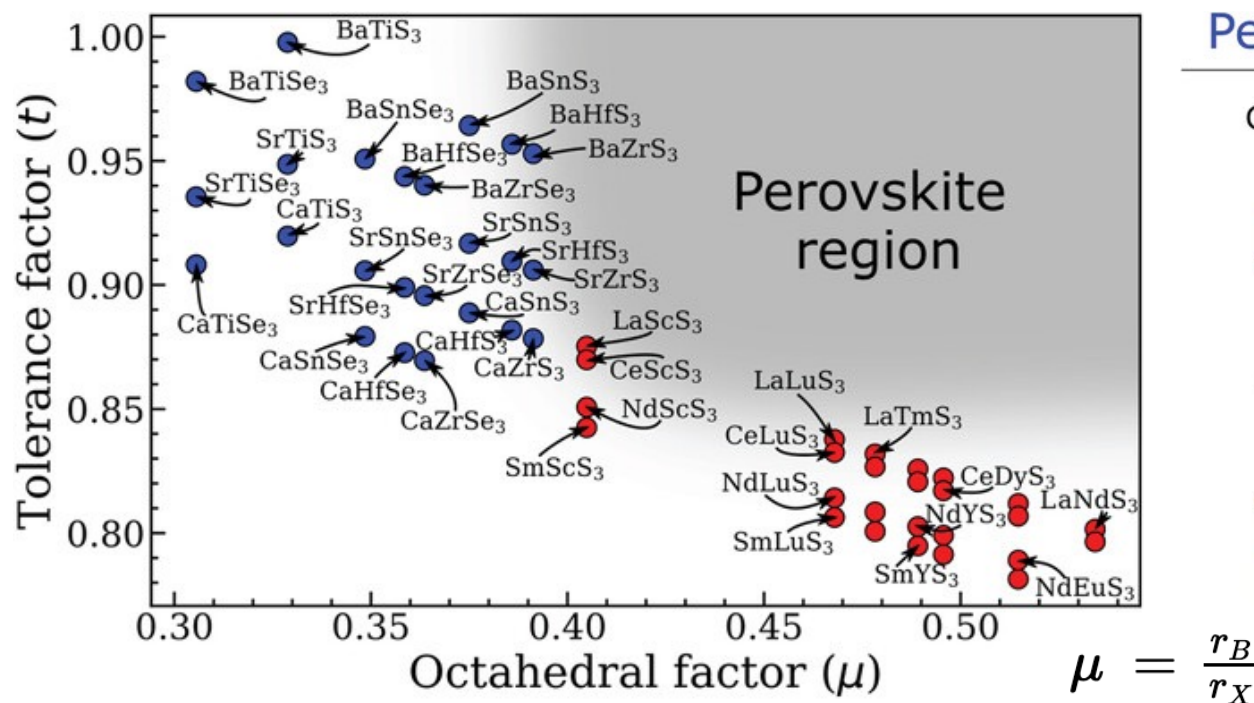
Marina R Filip
and Linn
Leppert
2024 *Electron.
Struct.* **6** 033002



Chalcogenide Perovskites

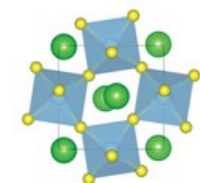
$$t = \frac{r_A + r_X}{\sqrt{2}(r_B + r_X)}$$

K. V. Sopiha et al. *Adv. Optical Mater.* 2022, 10, 2101704.

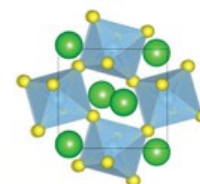


Perovskites

GdFeO₃-type

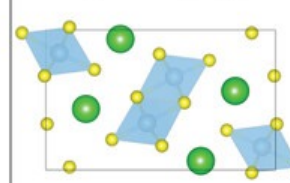


YScS₃-type

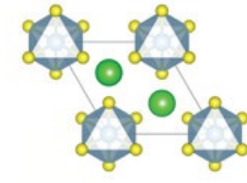


Non-perovskites

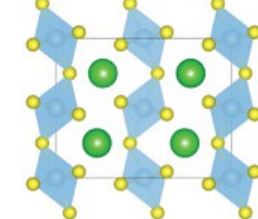
NH₄CdCl₃-type



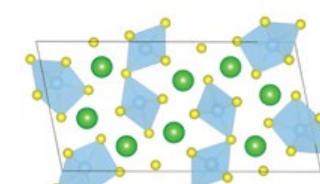
BaNiO₃-type



UFeS₃-type



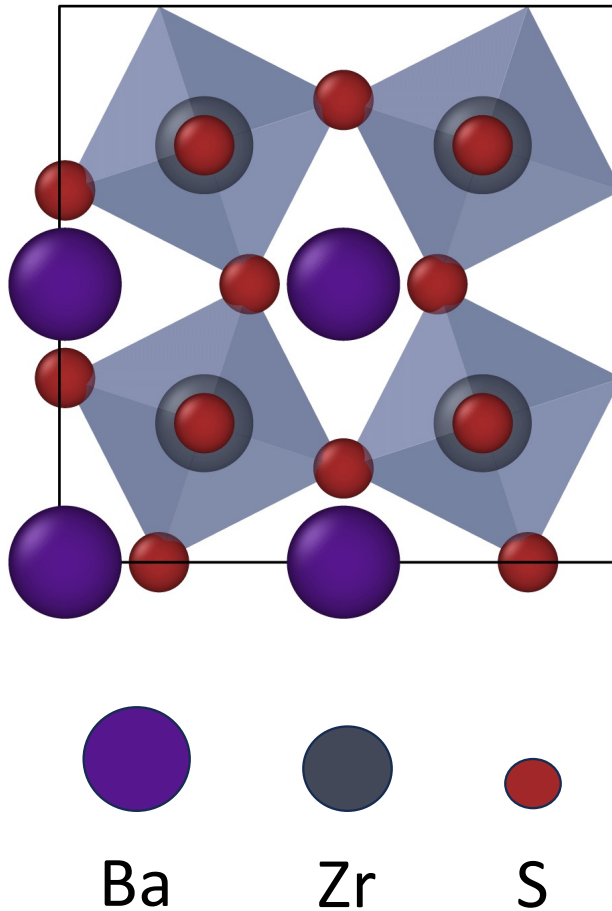
CeTmS₃-type



● - A ● - B ● - X

72 ABX₃ chalcogenide compounds reported in the experimental literature
25 of these form in the perovskite structure

BaZrS₃ (BZS): a tantalizing material



- Abundant and non-toxic
- Stable in air to 400°C
- Strong light absorption
- 1.8eV band gap → photovoltaics
- Tunable E_g through S/Se or Zr/Hf mixing
- Low thermal conductivity
(1.84 W/mK @ 300 K)

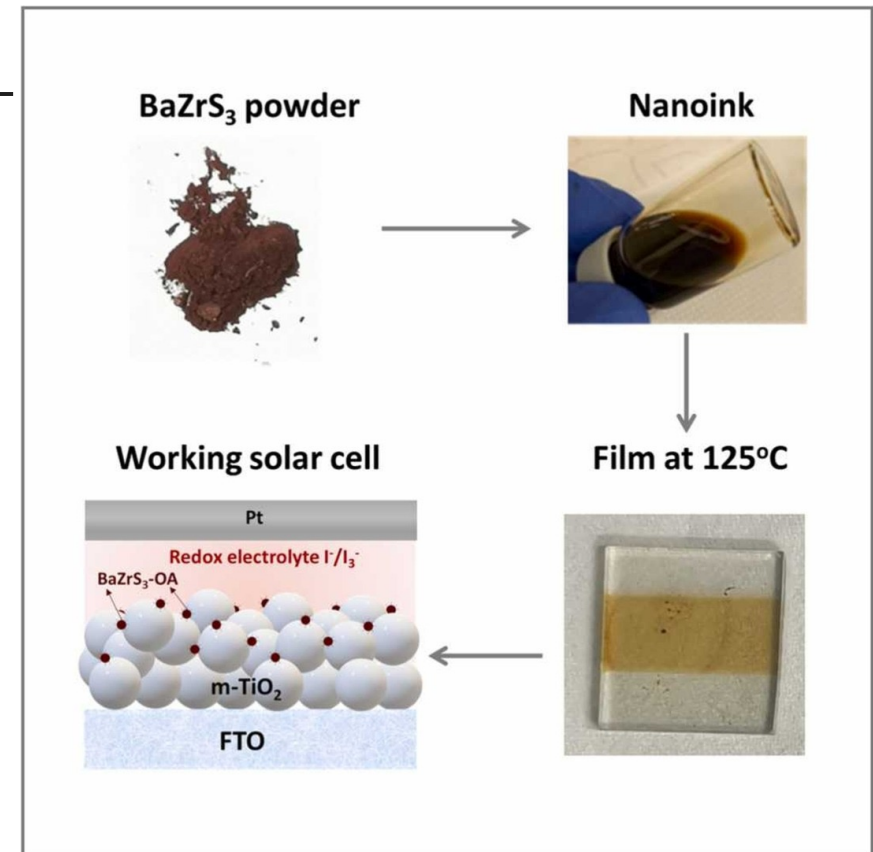
BaZrS₃ (BZS): an early-stage material

Volume 39, June 2024, 108608

Exploring the potential of powder-to-film processing for proof-of-concept BaZrS₃ perovskite solar cells

P. Dallas^a, K. Gkini^a, A. Kaltzoglou^c, L. Givalou^a, M. Konstantakou^a, S. Orfanoudakis^{a,d},
N. Boukos^a, E. Sakellis^a, P. Tsipas^a, A. Kalafatis^a, A.G. Karydas^b, A. Lagogiannis^b,
P. Falaras^a, V. Psycharis^a, T. Stergiopoulos^a ✉

Power conversion efficiency: 0.11%



Outline

Methods

- Predicting phase stability with quantum chemical simulations (is the material stable? under which conditions?)

Results

- BaZrS₃ synthesis
- BaZrS₃ polymorph stability
- Compositional engineering: BaZr(S,Se)₃
- Crystal structure engineering: Ba_{n+1}Zr_nS_{3n+1} Ruddlesden-Popper phases

Outlook

- What is required to establish chalcogenide perovskites as a new class of functional material?



Starting point: atoms move

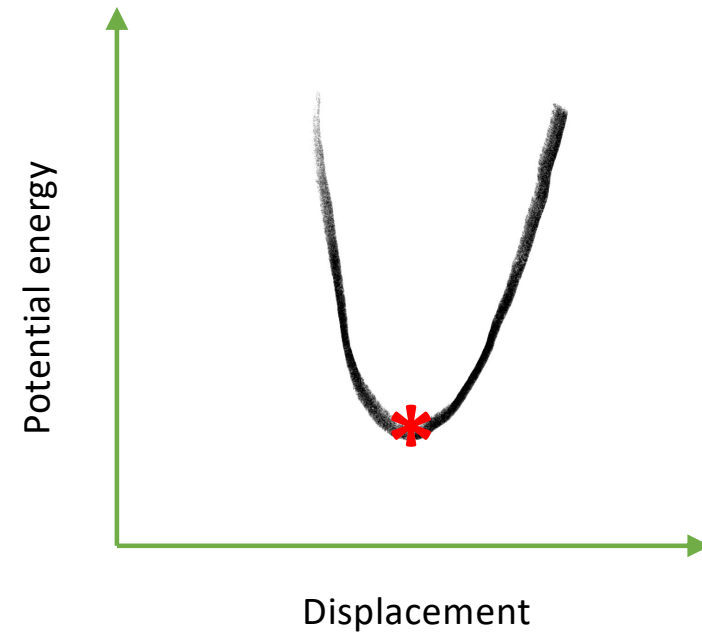
"A crystal is like a class of children arranged for drill, but standing at ease, so that while the class as a whole has regularity both in time and space, each individual child is a little fidgety!"

Dame Kathleen Lonsdale FRS

Small displacements from average position



Calm (low-energy) children

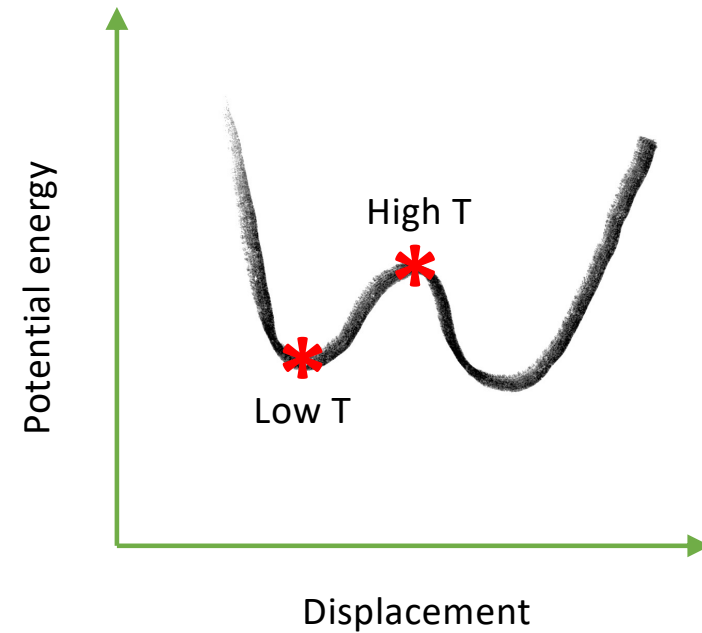


Harmonic approximation

Larger displacements from average position



Excited (high-energy) children



Crystal anharmonicity

Taylor expanding the potential energy surface

Crystal Potential
Static model

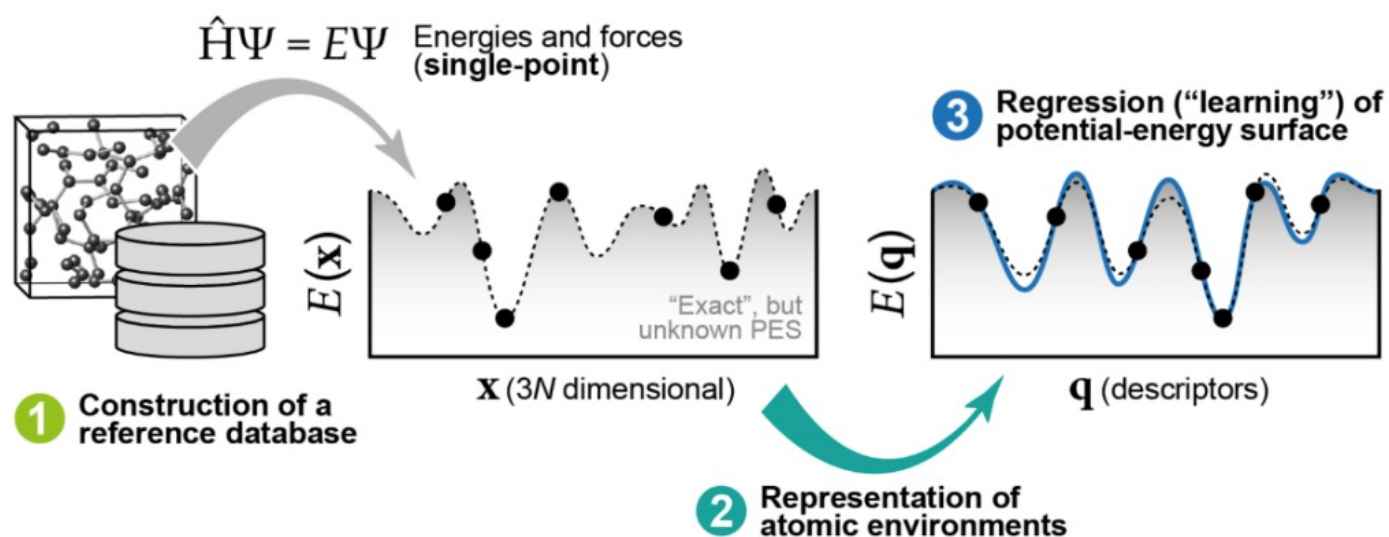
Harmonic Phonons
Non-interacting phonons
“Infinite lifetimes”

$$H = H_0 + \Phi_i^\alpha u_i^\alpha + \frac{1}{2} \Phi_{ij}^{\alpha\beta} u_i^\alpha u_j^\beta + \frac{1}{6} \Phi_{ijk}^{\alpha\beta\gamma} u_i^\alpha u_j^\beta u_k^\gamma + \dots$$

Ionic Forces
= 0 at equilibrium

Anharmonicity
Phonon scattering
Required for e.g. thermal conductivity

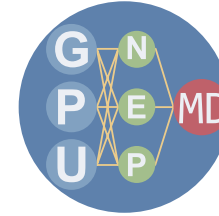
Machine-learning the potential energy surface



Machine Learning Interatomic Potentials as Emerging Tools for Materials Science
Volker L. Deringer, Miguel A. Caro, Gábor Csányi
Adv. Mater. 2019, 31, 1902765

Neuroevolution Potential (NEP)

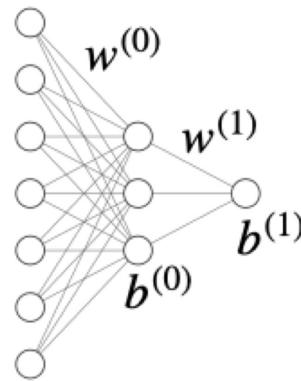
Fan *et al* Phys. Rev. B 2021, 104, 104309



Machine Learned
Interatomic
Potential / MD



Density Functional
Theory
HSE06 XC-functional



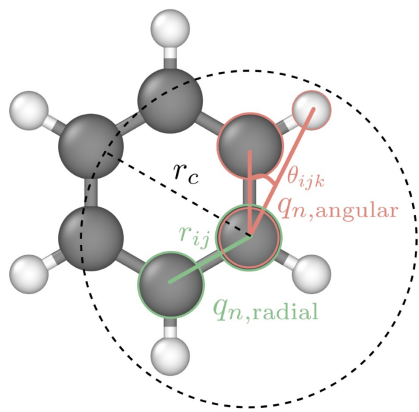
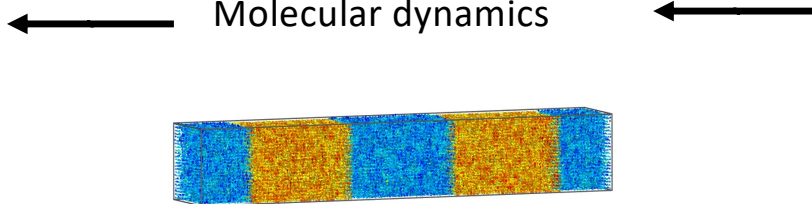
Input features

Output

$$E_i = \sum_{\mu=1}^{N_{neu}} w_{\mu}^{(1)} \tanh \left(\sum_{\nu=1}^{N_{des}} w_{\mu\nu}^{(0)} q_{\nu}^i - b_{\mu}^{(0)} \right) - b^{(1)}$$

$$F_i = \sum_{j \neq i} \frac{\partial E_i}{\partial r_{ij}} - \frac{\partial E_j}{\partial r_{ji}} \quad \mathbf{w}_i = \sum_{j \neq i} \mathbf{r}_{ij} \otimes \frac{\partial E_j}{\partial \mathbf{r}_{ji}}$$

Molecular dynamics



Results I: BaZrS₃ synthesis

Challenge: moderate temperature synthesis

“Chalcogenide perovskite synthesis is pretty damn hard”

- Jonathan Scragg, MRS Fall 2022

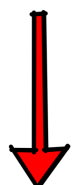
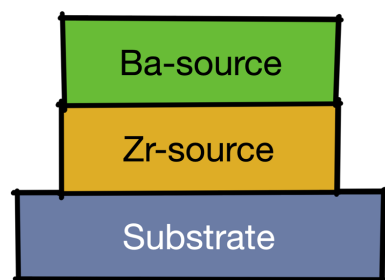
"It works if you're a good solid state chemist and you don't mind breaking ampoules....its messy”

- Thomas Stergiopolous, MATSUS 2025

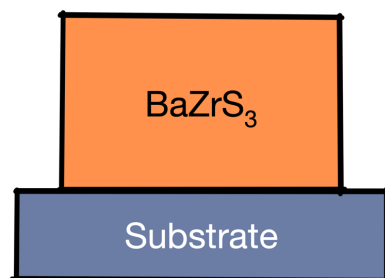
"We have tonnes of experiments which do not work”

- Thomas Stergiopolous, MATSUS 2025

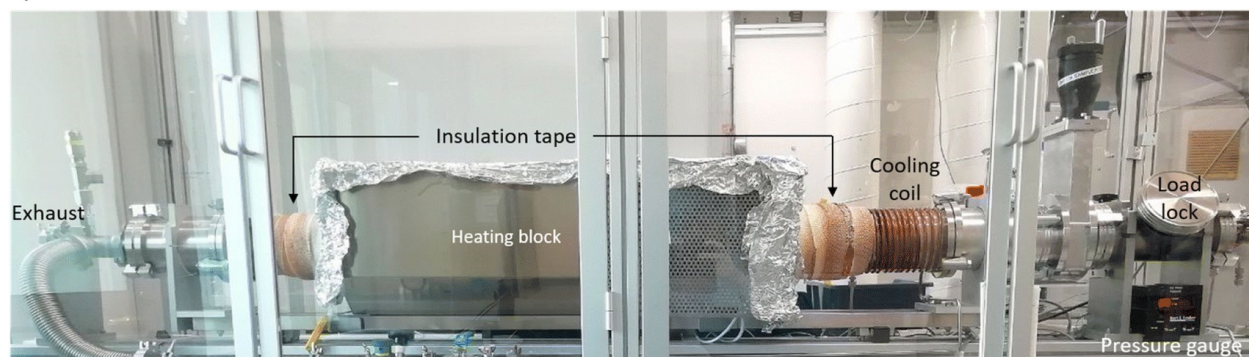
Challenge: moderate temperature synthesis



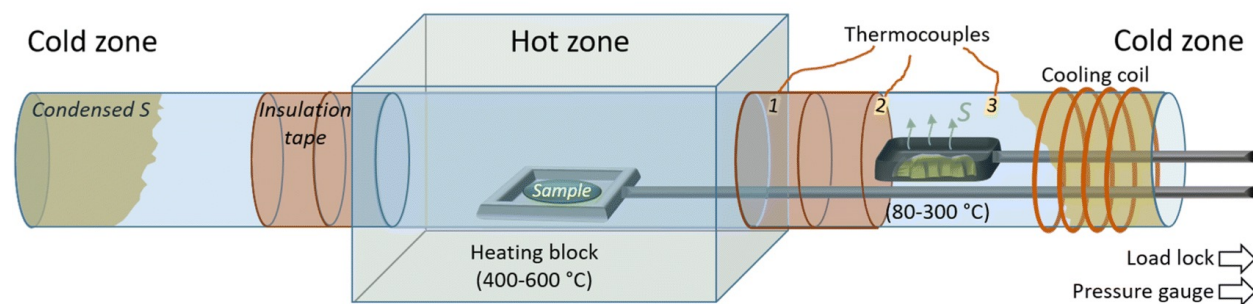
Sulfur
Annealing



a)



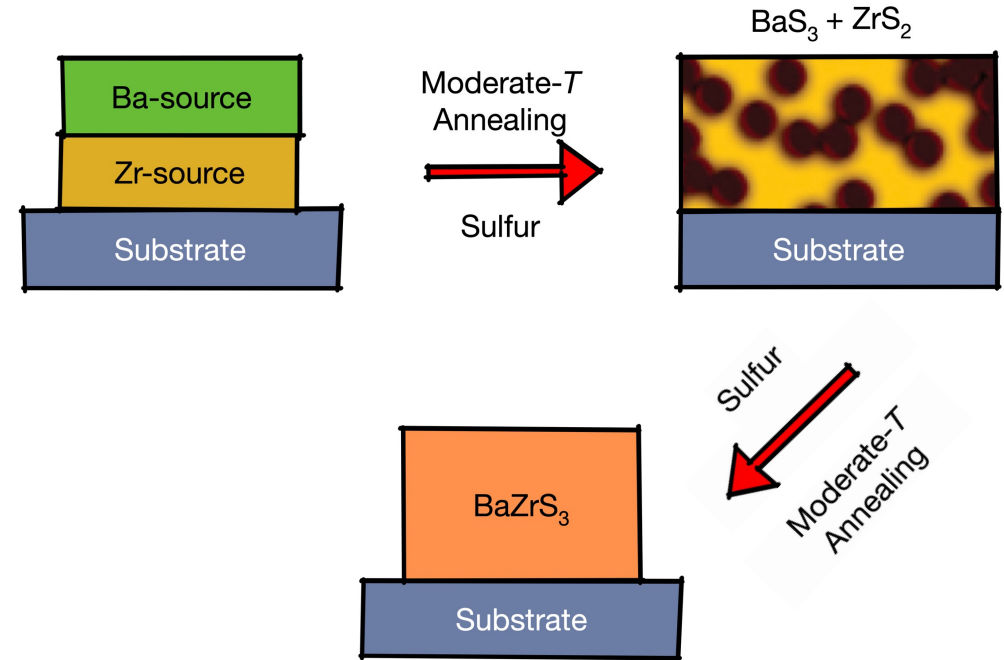
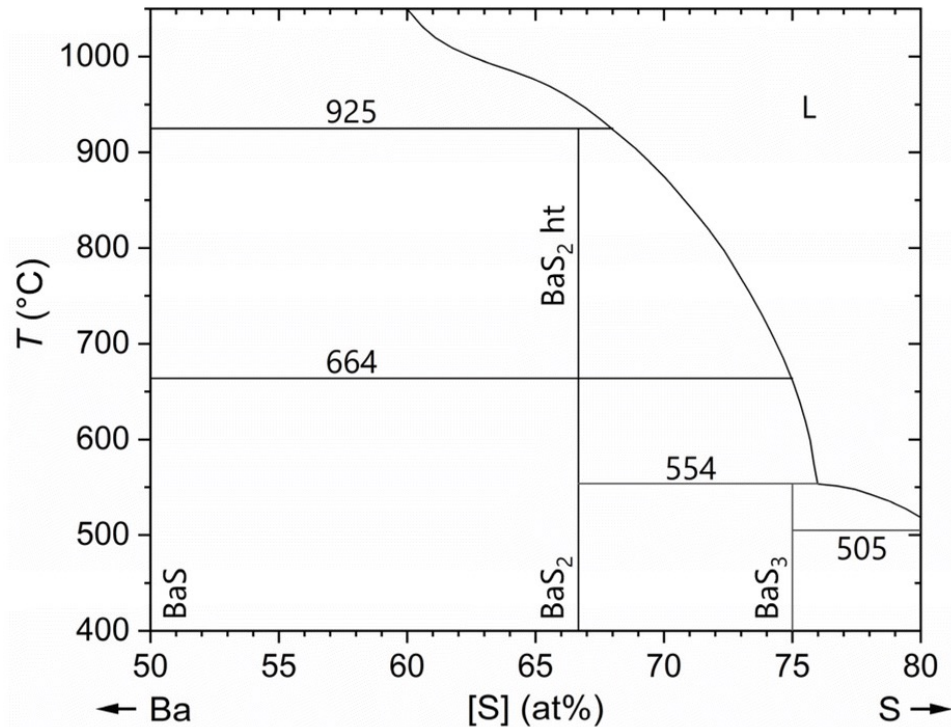
b)



Schematic of specialised furnass for chalcogenide
perovskite synthesis

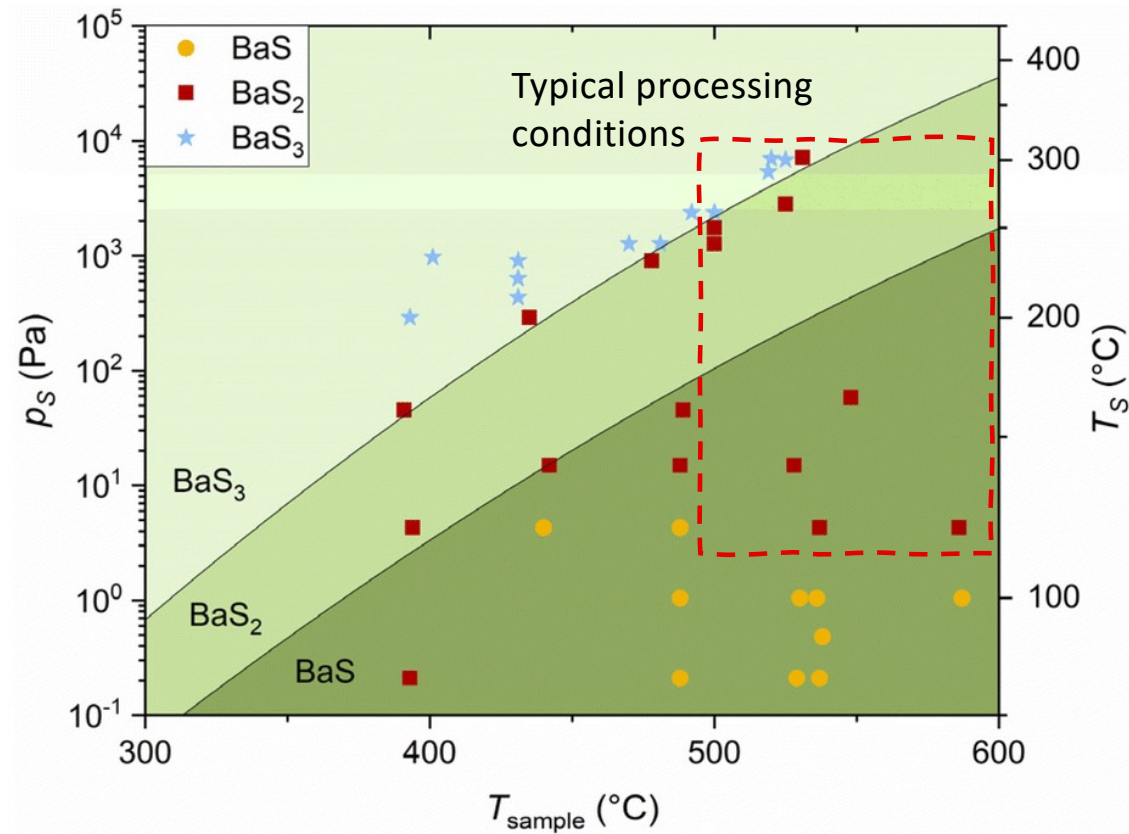
Comparotto, Whalley, Sopiha et al
J. Mater. Chem. A, 2025,13, 9983-9991

BaS₃ intermediate to form liquid flux and overcome kinetic barriers



Digitised from K. C. Vincent , S. Agarwal , J. W. Turnley and R. Agrawal , Adv. Energy Sustainability Res., 2023, 4 , 2300010

p_S - T phase diagram for the Ba-S system

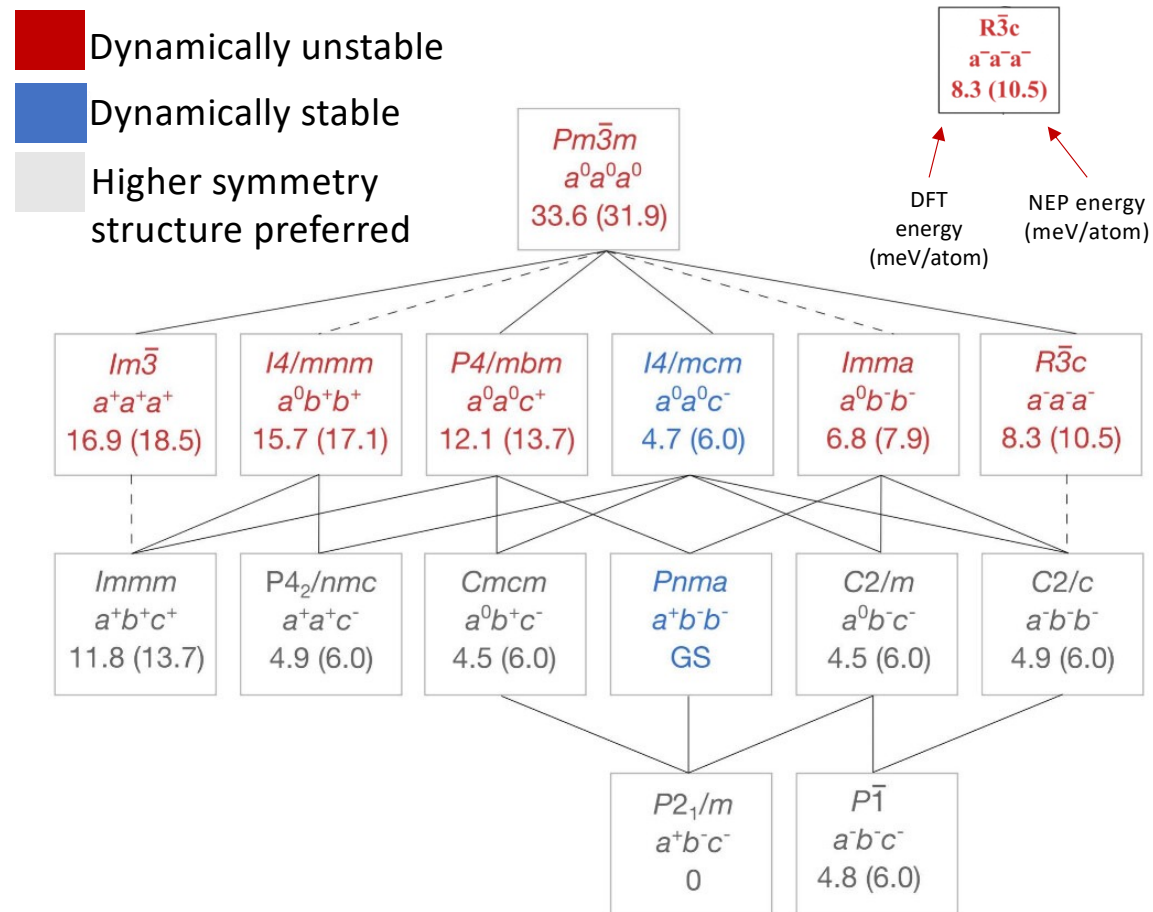
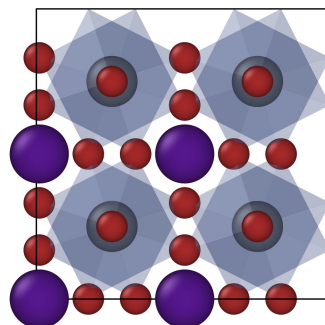
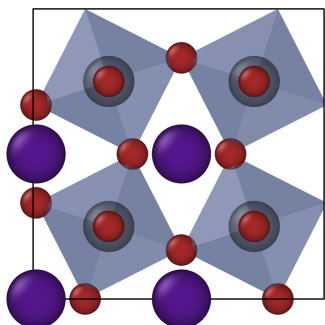
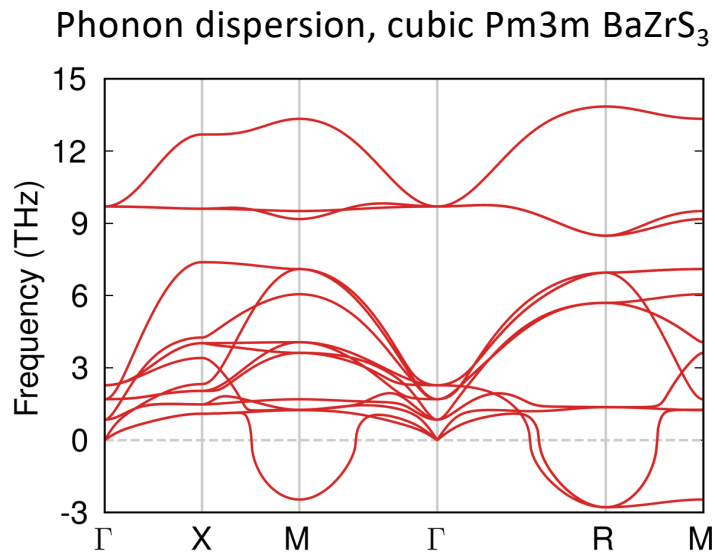


Comparotto, Whalley, Sopiha et al
J. Mater. Chem. A, 2025,13, 9983-9991

High sulfur pressures ($>10^3$ Pa at 500 $^{\circ}\text{C}$) are required to form BaS_3 .

Results II: BaZrS₃ polymorph stability

Which BaZrS₃ perovskite polymorphs are formed?



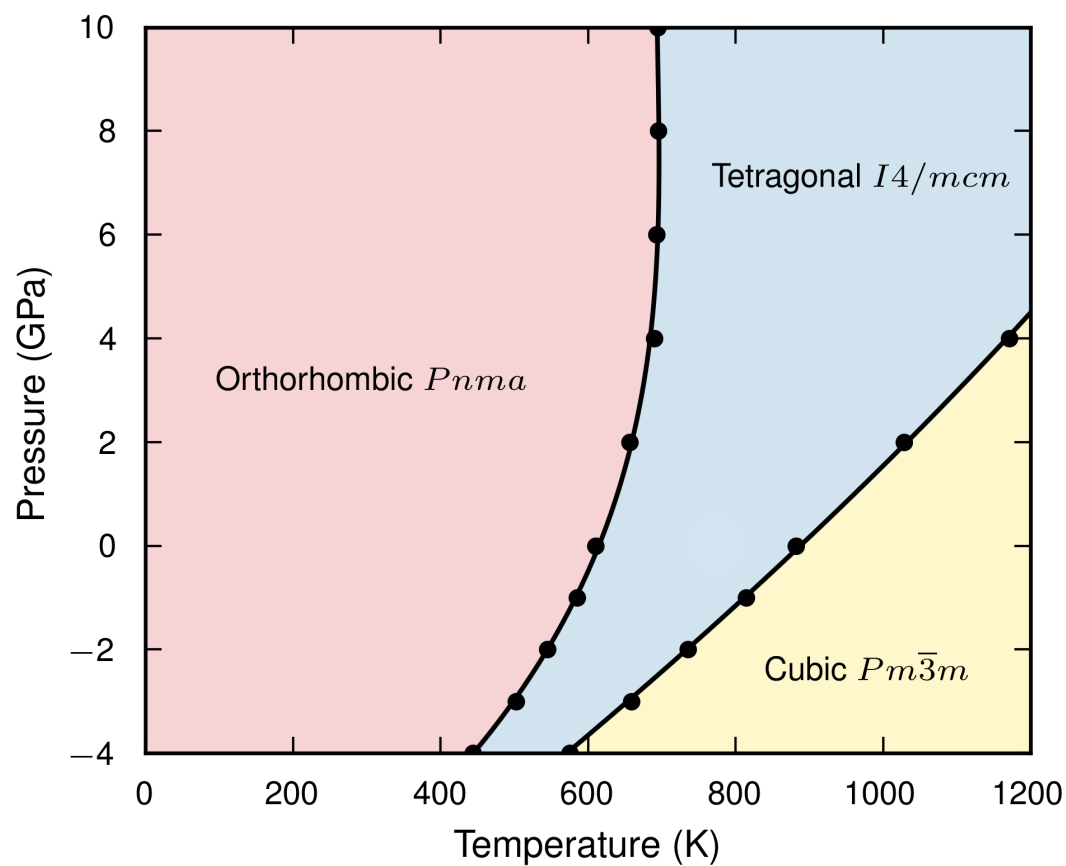
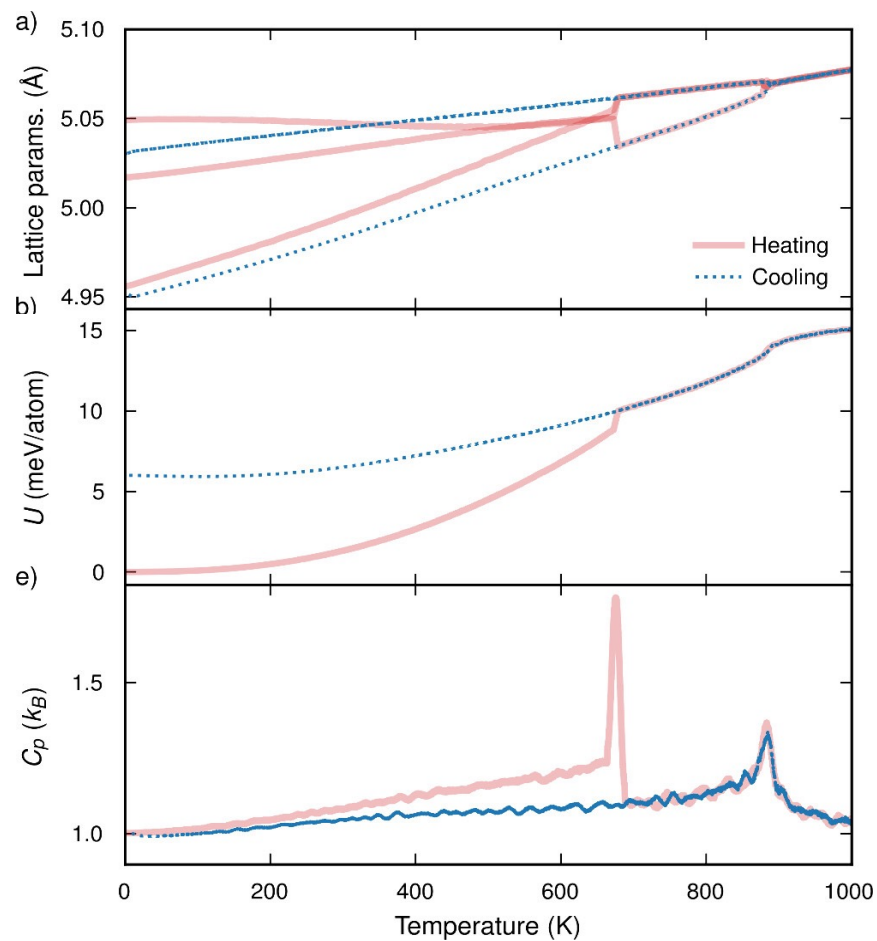
Adapted from *Group-Theoretical Analysis of Octahedral Tilting in Perovskites*, C. J. Howard and H. T. Stokes

BaZrS₃ phase diagram

Calculated through NEP-MD and thermodynamic integration (~40,000 atoms)

P. Kayastha, E. Fransson, P. Erhart, L. Whalley

Phys. Chem. Lett. 2025, 16, 8, 2064–2071

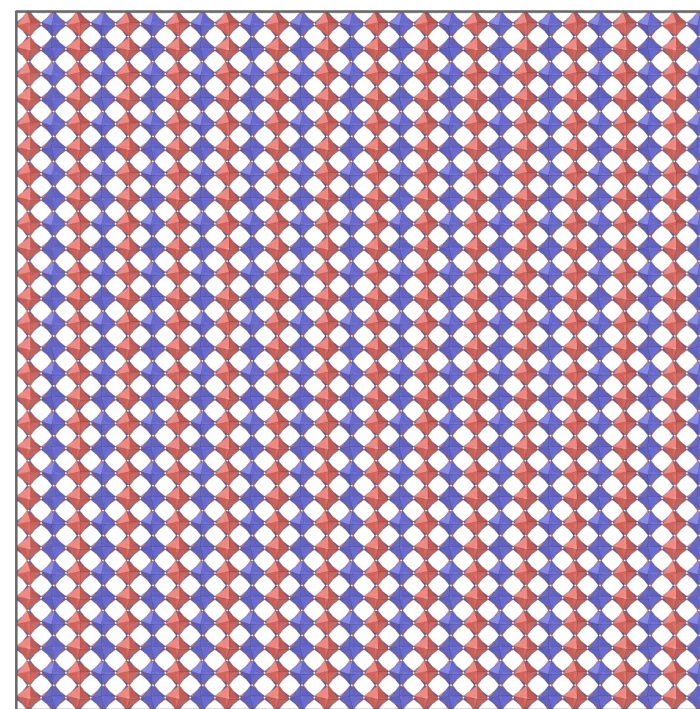
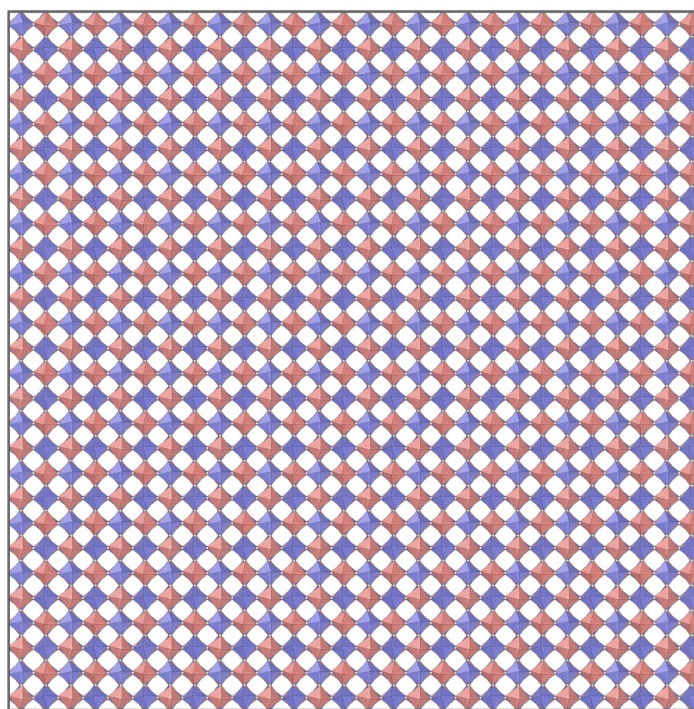
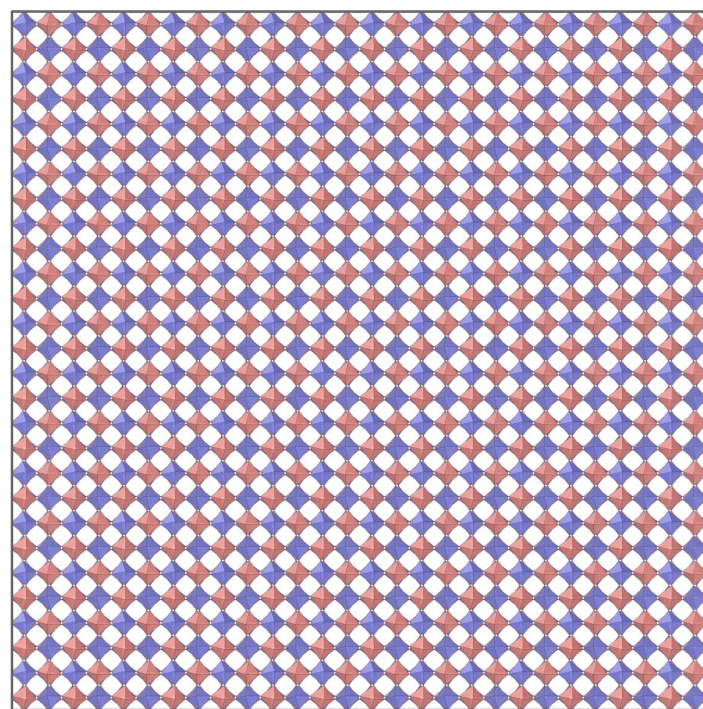


BaZrS₃ octahedral tilt phase transitions

P. Kayastha, E. Fransson, P. Erhart, L. Whalley
Phys. Chem. Lett. 2025, 16, 8, 2064–2071

Video courtesy of Erik Fransson

T=1K

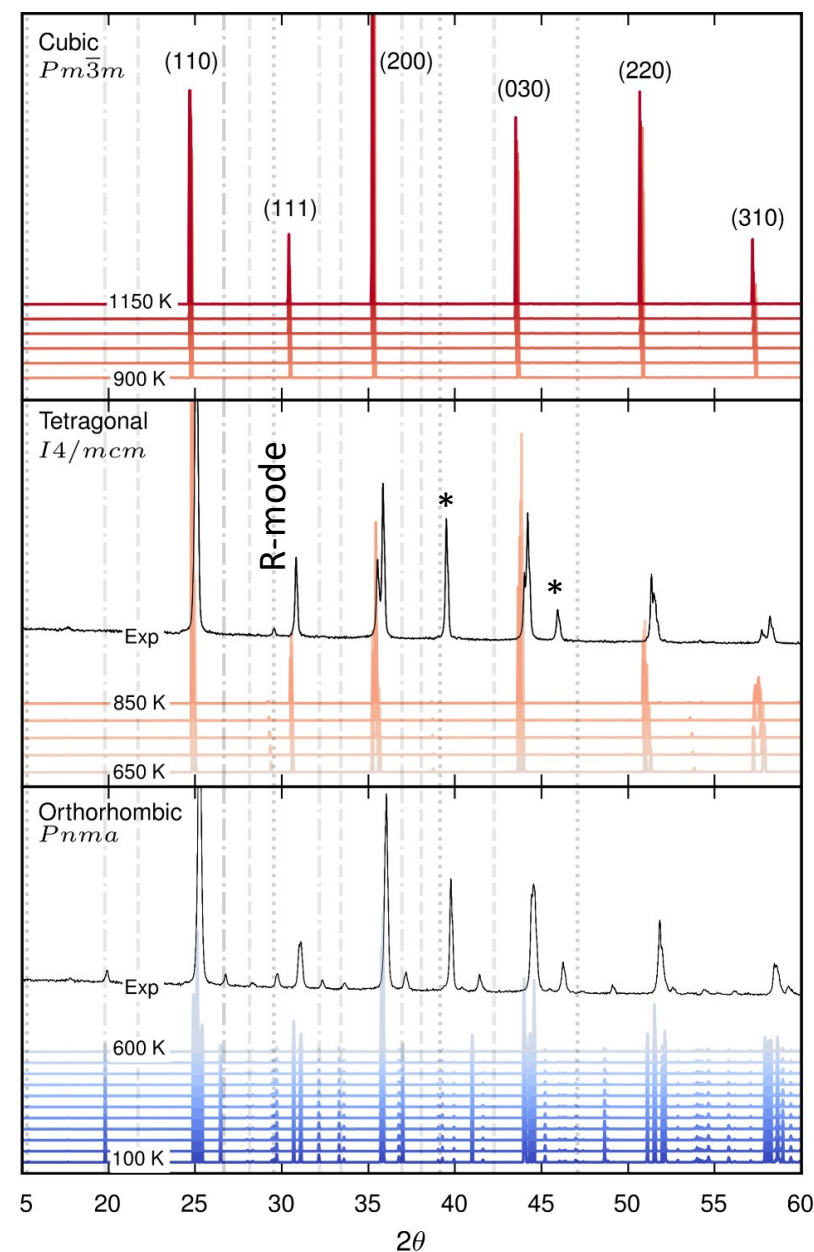


Predicted experimental signatures

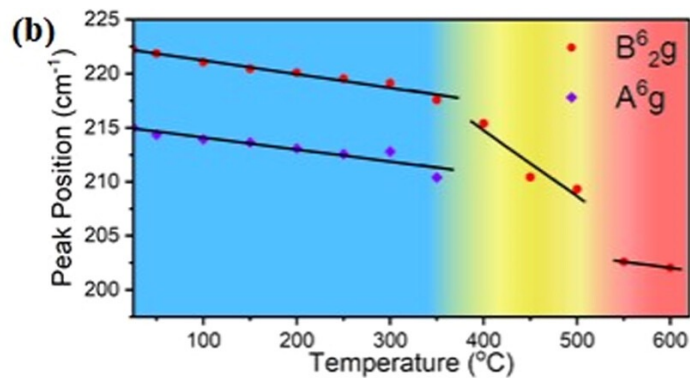
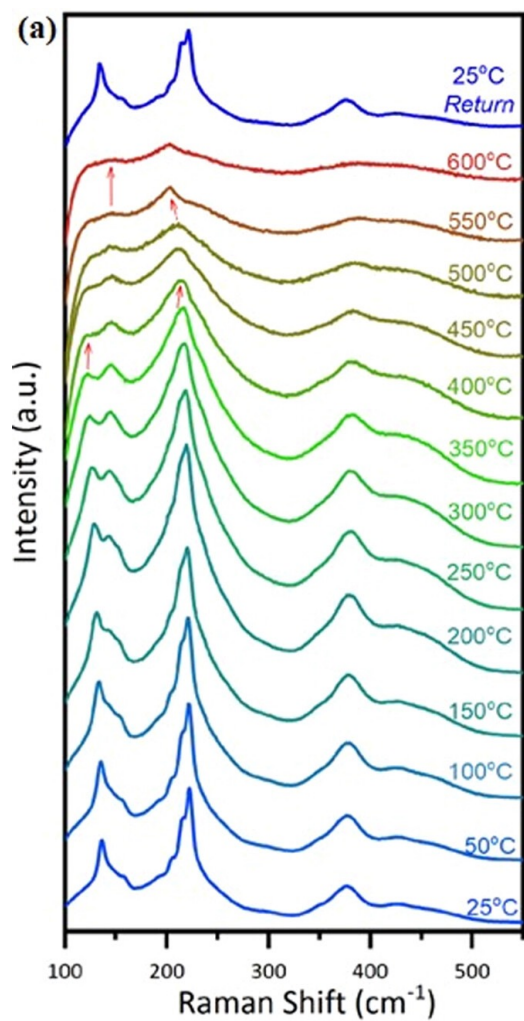
P. Kayastha, E. Fransson, P. Erhart and L. Whalley, Phys. Chem. Lett. 2025, 16, 8, 2064–2071

- The R-, M-, and X-point distortions are represented with dotted, dashed, and dash-dotted lines, respectively.
- Tetragonal: R-mode activation at 29°->, peak splitting
- Cubic: M-mode activation at 27° and 33°, further peak splitting.
- * are associated with Pt strip used to heat sample

Experimental data from
Bystricky et al, Inorg. Chem.
(2024) **63** 12826

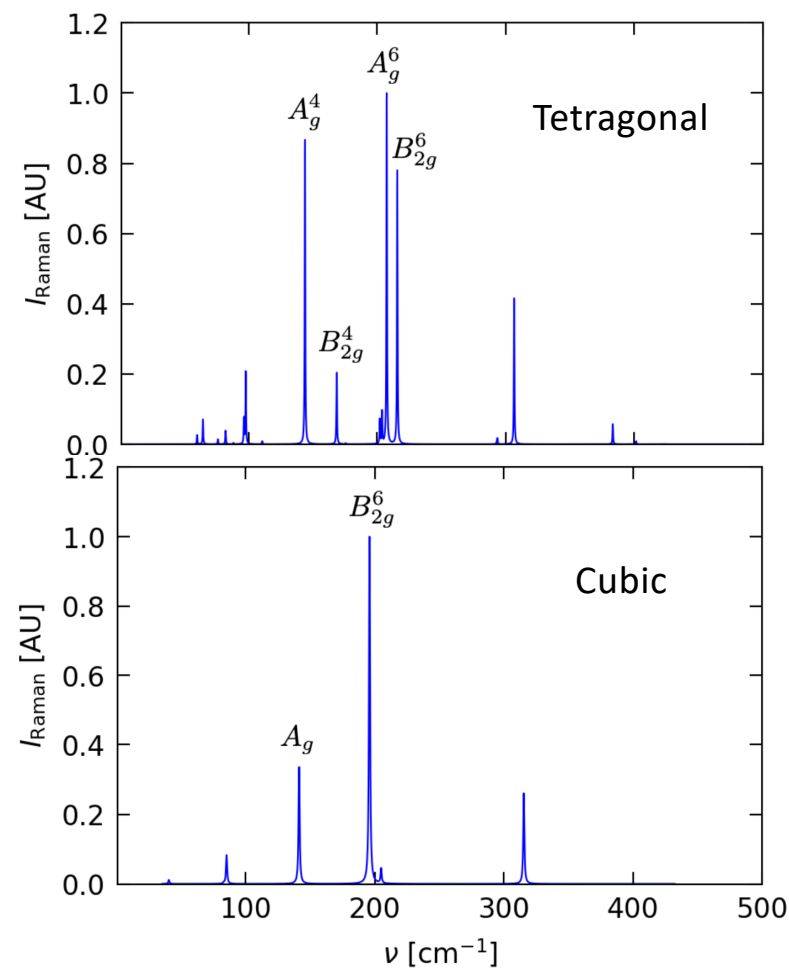


Experimental Raman spectra



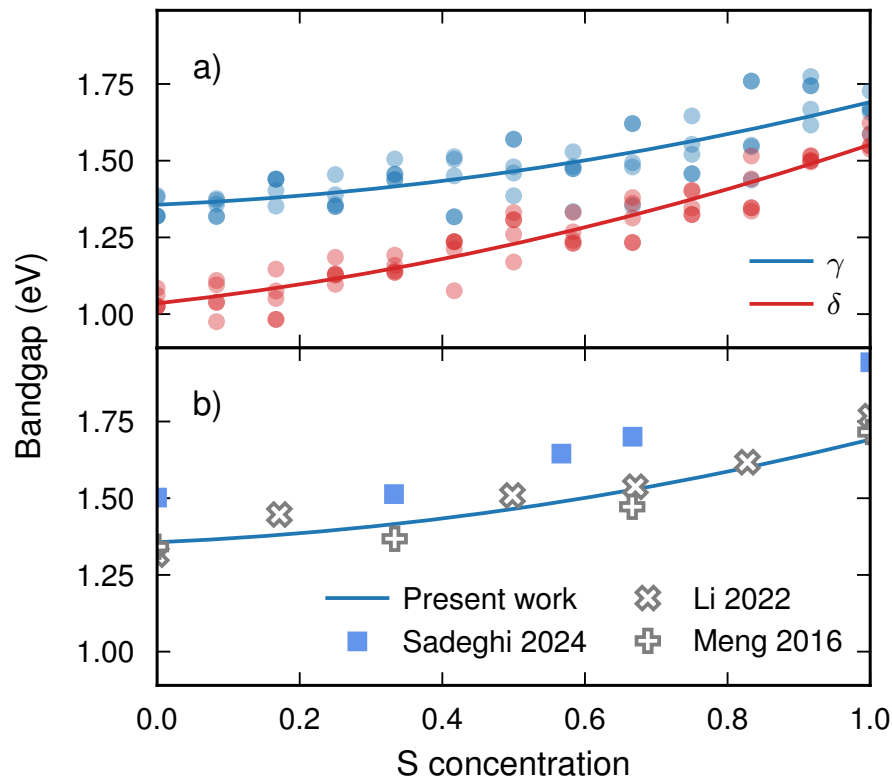
Jaiswal et al, Inorg. Chem. 2024, 63, 51, 24157–24166

Predicted first-order Raman spectra

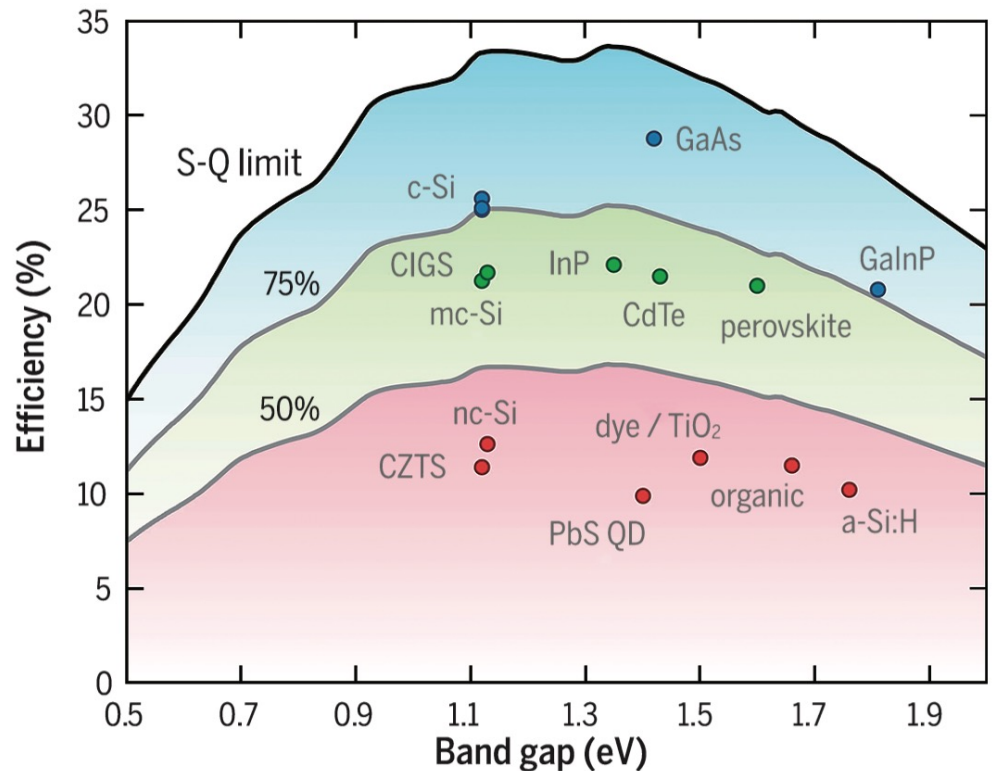


Results III: Compositional engineering, $\text{BaZr}(\text{S,Se})_3$

Incorporating Se reduces the band gap towards the optimal Shockley-Queisser value.

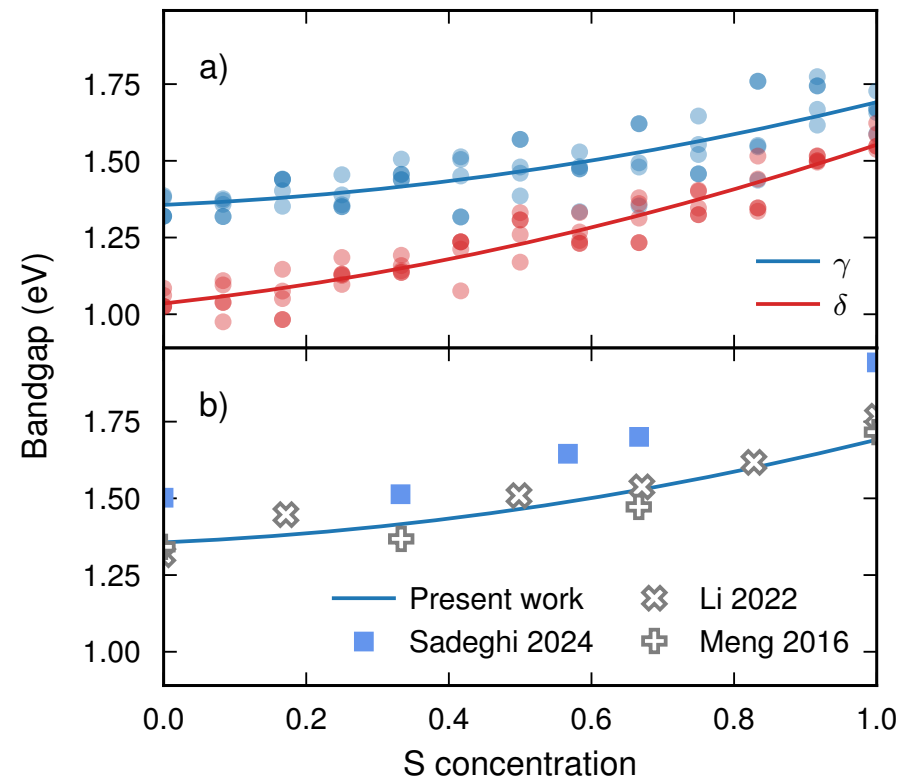
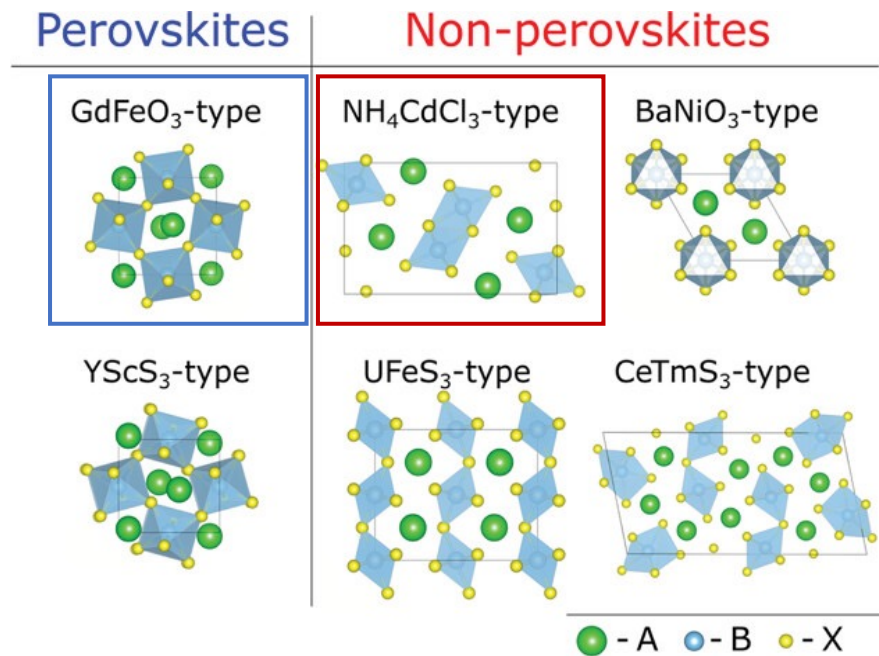


E. Fransson, P. Kayastha,, P. Erhart and L. Whalley, In Prep.



Polman et al., Science 352, aad4424, (2016)

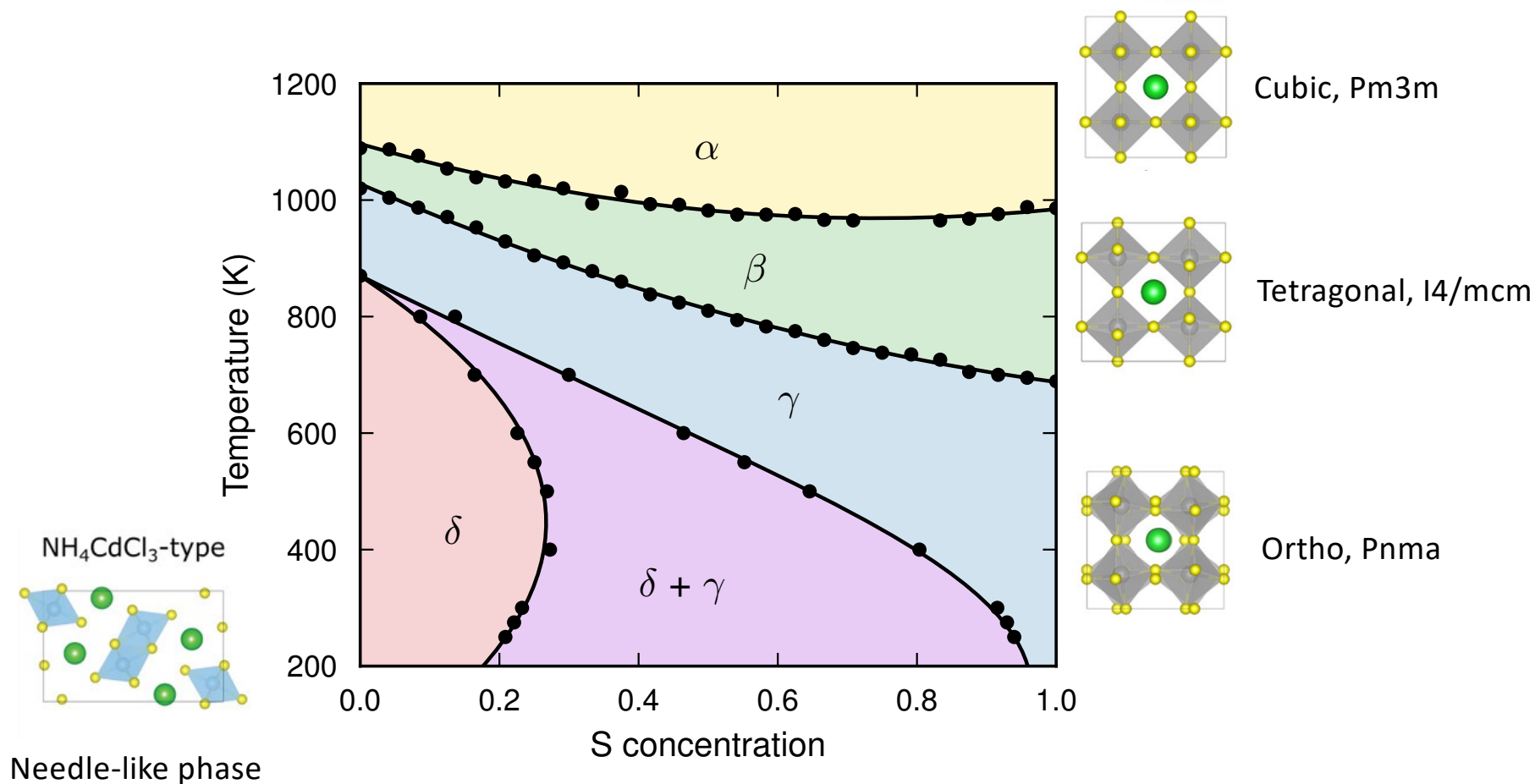
Incorporating Se reduces the band gap towards the optimal Shockley-Quiesser value.



BaZr(S,Se)₃ phase diagram

Calculated through NEP-MD and thermodynamic integration

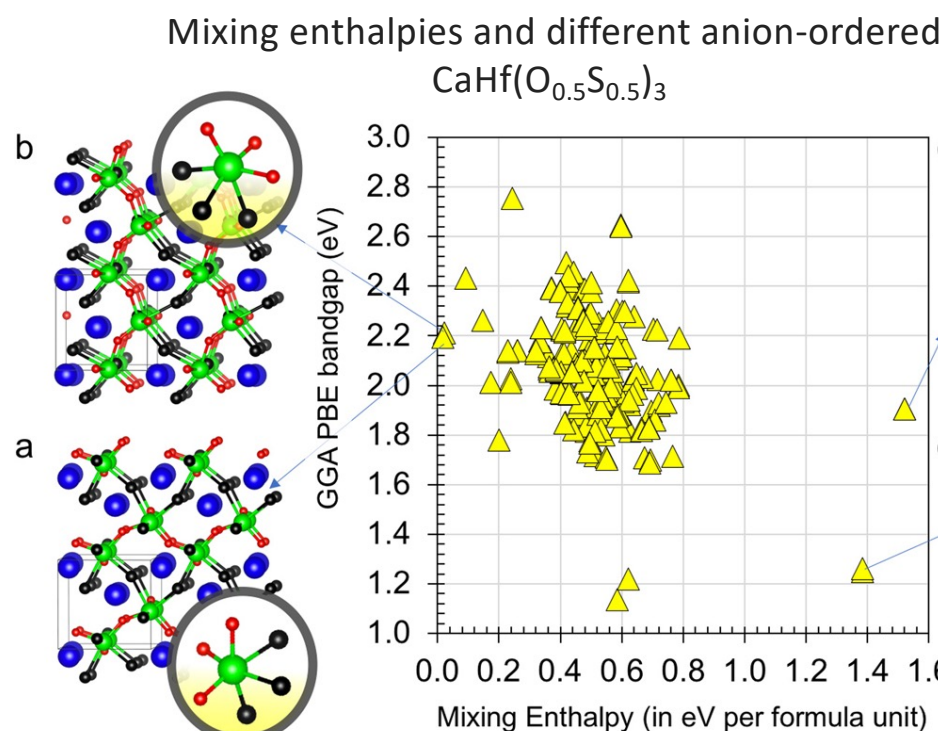
E. Fransson, P. Kayastha,, P. Erhart
and L. Whalley, In Prep.



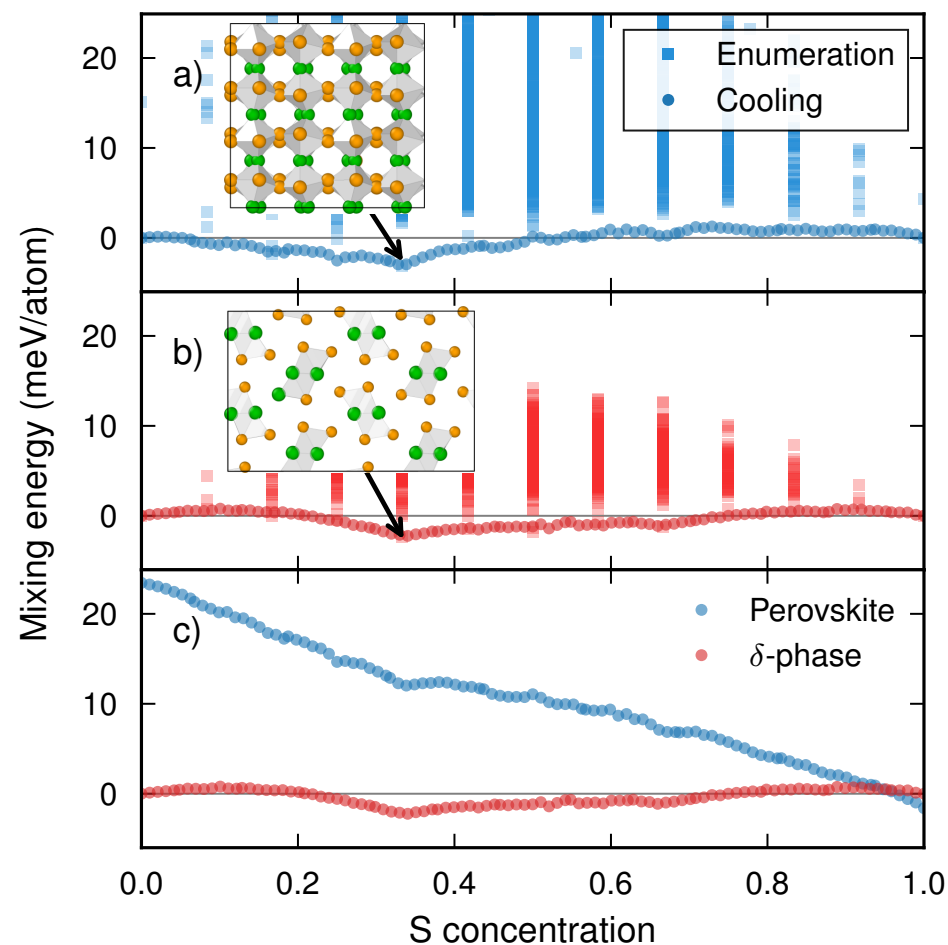
Anion ordering in $\text{BaZr}(\text{S},\text{Se})_3$

Calculated through NEP-MD

E. Fransson, P. Kayastha,, P. Erhart
and L. Whalley, In Prep.



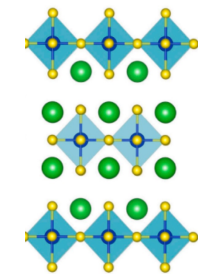
Pilania, G., Ghosh, A., Hartman, S.T. *et al.*
npj Comput Mater **6**, 71 (2020)



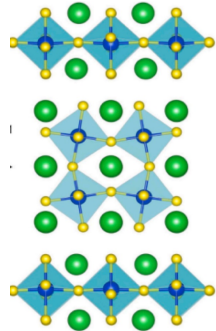
Results IV: Crystal structure engineering,
 $\text{Ba}_{n+1}\text{Zr}_n\text{S}_{3n+1}$ Ruddlesden-Popper phases

Formation of Ruddlesden-Popper materials $\text{Ba}_{n+1}\text{Zr}_n\text{S}_{3n+1}$

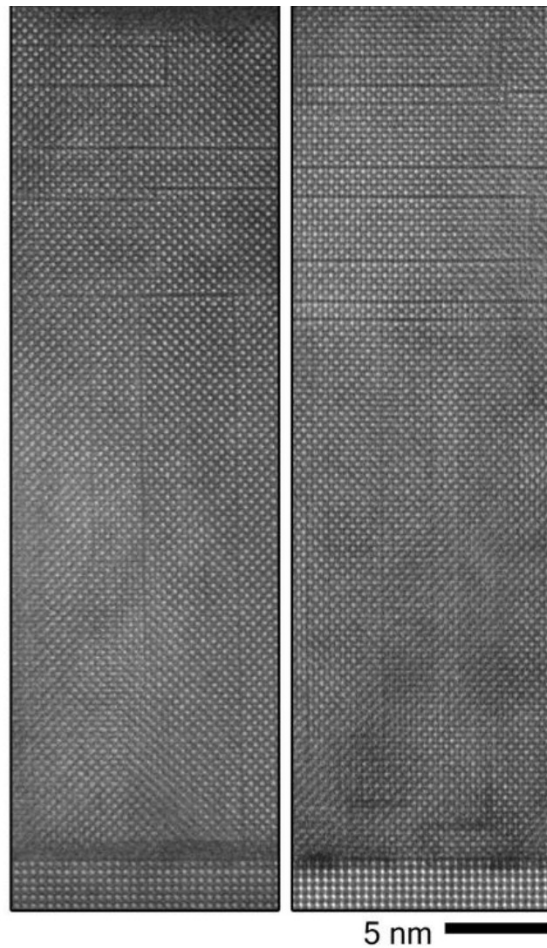
$\text{BaZr}(\text{S},\text{Se})_3$ films with RP faults



$n=1$

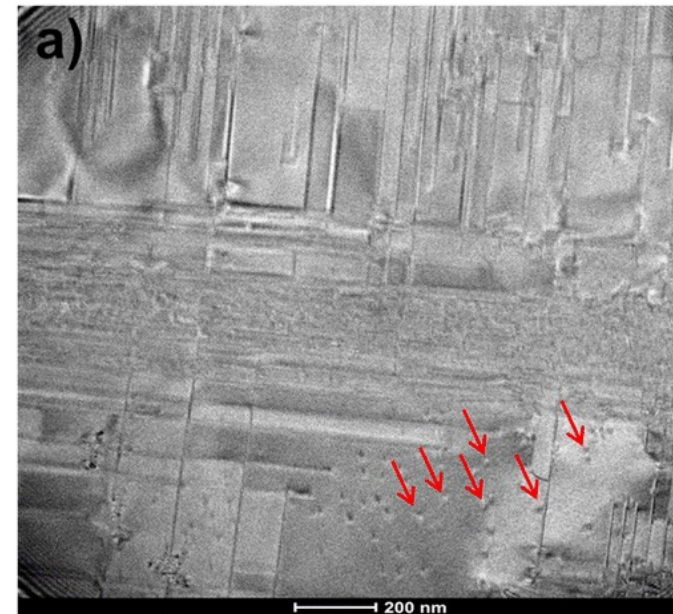


$n=2$



5 nm

ZrO_2 acting as a Zr-sink

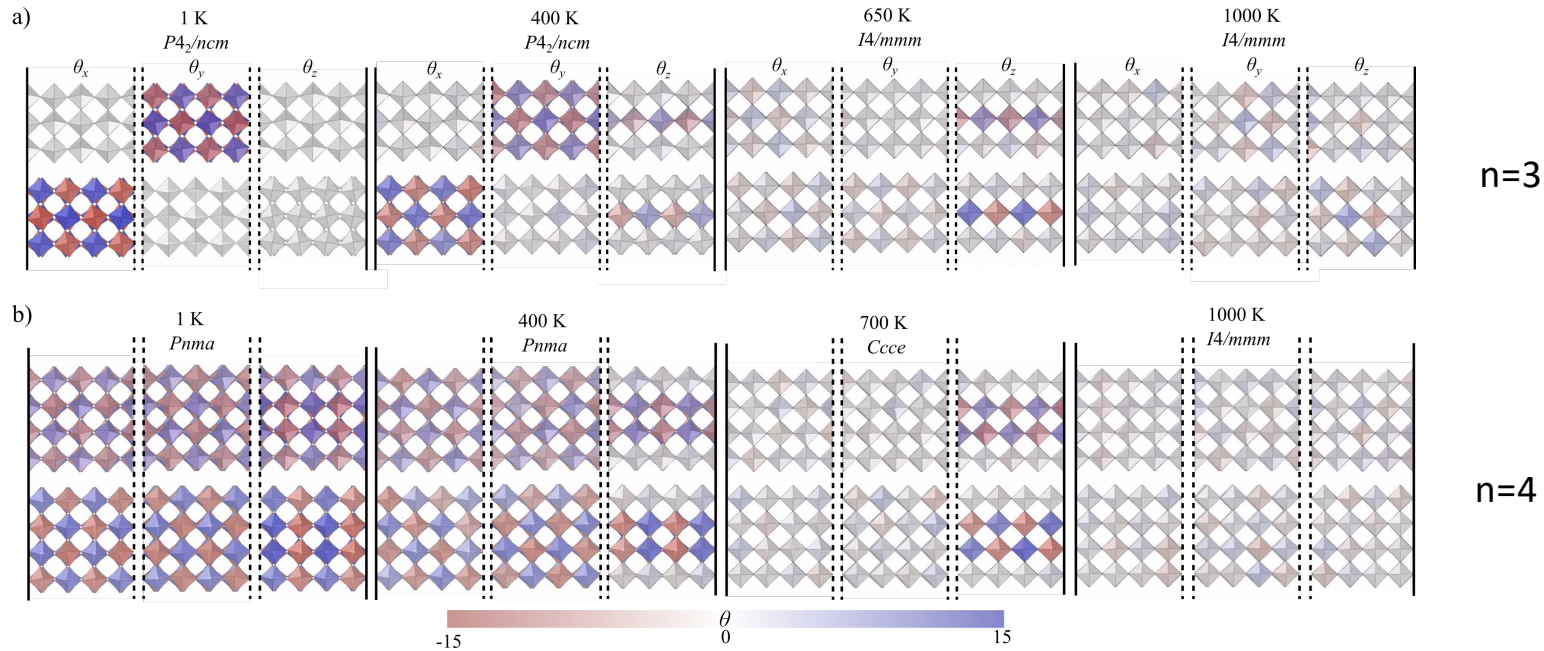
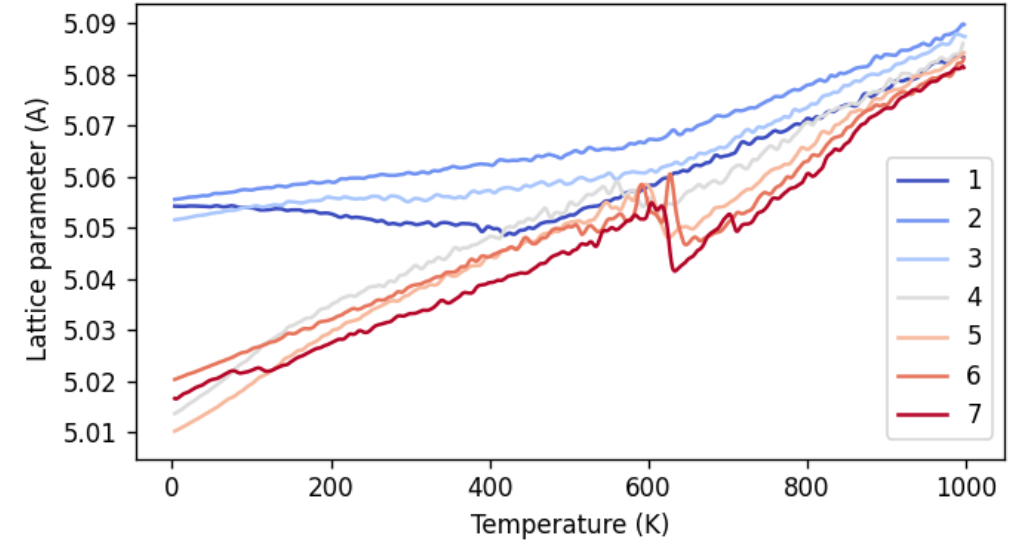


Materials Chemistry Frontiers
2024,8, 3358-3372

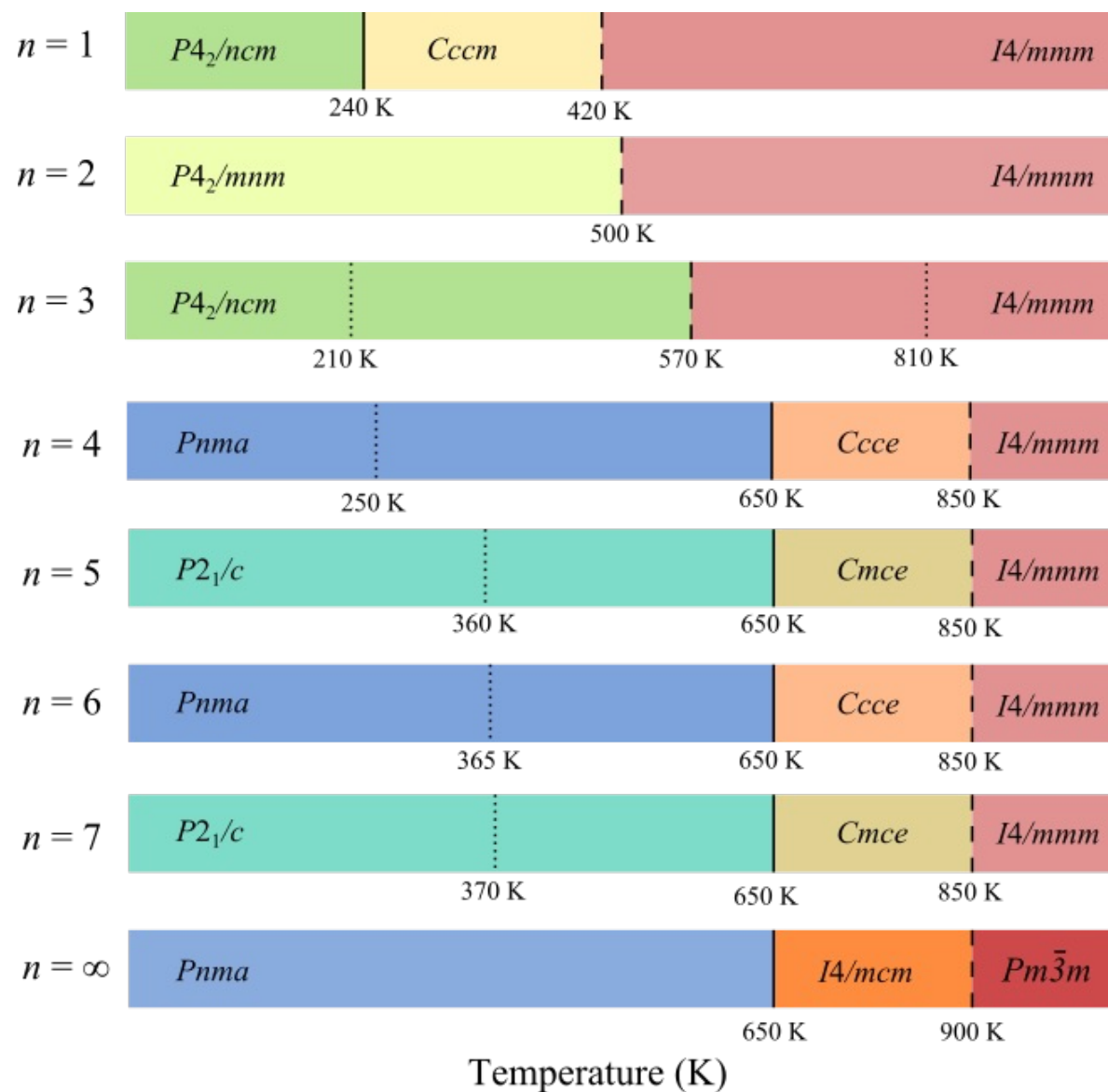
Sadeghi et al, Advanced
Functional Materials,
2023, 33(41)

Ruddlesden Popper phase transitions, $n=1$ to $n=7$

P. Kayastha, E. Fransson,, P. Erhart
and L. Whalley, In Prep.



Ba_{n+1}Zr_nS_{3n+1} phase transitions n=1 to n=7



P. Kayastha, E. Fransson, P. Erhart
and L. Whalley, In Prep.

Outstanding challenges for CPV development

- 1) Preventing formation of Zr binary phases → off-stoichiometry
- 2) Controlling Ruddlesden-Popper phase formation
- 3) Impact of RP & polymorphs on optoelectronic & thermal prop.
- 4) Alloying and doping strategies to tune properties
- 5) Lack of PL: identifying recombination pathways

Recommended Reading:

K. V. Sopiha, C. Comparotto, J. A. Márquez, J. J. S. Scragg,

Chalcogenide Perovskites: Tantalizing Prospects, Challenging Materials. *Adv. Optical Mater.* 2022, 10, 2101704. <https://doi.org/10.1002/adom.202101704>

Summary

- 1) Chalcogenide perovskites are a promising class of materials
- 2) Challenges around synthesis and perovskite phase stability
- 3) BaS_3 precursor requires high P_s ($>10^3$ Pa at 500 °C)
- 4) We predict a series of perovskite polymorph phase transitions
- 5) RP phases undergo a similar series of transitions, converging towards 3D perovskite behaviour for $n=4$
- 6) $\text{BaZr}(\text{S},\text{Se})_3$ perovskite is thermo. unstable for Se content $> 10\%$

Collaborators



Octahedral Tilting
at Chalmers



Prakriti Kayastha



Jonathan Scragg



Corrado Comparotto



Erik Fransson



Paul Erhart



Giulia Longo

Thank you

Lucy Whalley

l.whalley@northumbria.ac.uk

